

PAPER_352 R Aquarii Symbiotic Binary: F_U_Bi_i with Kepler 44-Year Orbital Period

Whitepaper Series: Star-Magic UQFF Phase 2

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Classification: FIRST UQFF F_U_Bi_i for a symbiotic nova binary with P_orb = 44 yr

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Abstract

R Aquarii is the nearest symbiotic nova system, consisting of a Mira Pulsating Giant and a white dwarf companion in a 44-year orbit. UQFF buoyancy-unified force F_U_Bi_i - 2.09×10^{12} N is calculated at the binary orbital radius derived from Kepler's third law: $a_{\text{orb}} = (GM_{\text{total}}P_{\text{orb}}^2/4\pi^2)^{1/3}$. HST 2025 near-UV imaging of the expanding jet system (launched ~10 yr ago) provides the observational anchor. The UQFF Kozima LENR coupling is relevant at the mass transfer interface between the giant's wind and the WD accretion disk.

2. Core Physics

2.1 UQFF Buoyancy-Unified Force

$$F_{U_Bi_i} \approx -2.09 \times 10^{12} \text{ N}$$

(intermediate between TDE-scale PAPER_351 and AGN-scale PAPER_346)

2.2 Kepler's Third Law Orbital Radius

$$a_{\text{orb}} = \left(\frac{GM_{\text{total}}P_{\text{orb}}^2}{4\pi^2} \right)^{1/3}$$

with P_orb = 44 yr = 1.388×10^9 s and M_total = M_Mira + M_WD $\approx 2 M_{\odot}$:

$$a_{\text{orb}} = \left(\frac{6.674 \times 10^{-11} \times 4 \times 10^{30} \times (1.388 \times 10^9)^2}{4\pi^2} \right)^{1/3} \approx 1.0 \times 10^{13} \text{ m} \approx 70 \text{ AU}$$

2.3 Mass Transfer and LENR Coupling

At the tidal mass transfer interface:

$$\dot{M}_{\text{transfer}} = \dot{M}_{\text{wind}} \cdot \left(\frac{a_{\text{orb}}}{R_{\text{Mira}}} \right)^{-2}$$

The UQFF Kozima LENR force:

$$F_{\text{Kozima}} = E_{\text{LENR}}/a_{\text{orb}}$$

where E_LEN is the low-energy nuclear reaction energy scale at the compressed accretion interface.

2.4 HST 2025 Jet Observation

R Aquarii's expanding jets, first resolved by HST in the 1990s and re-observed in 2025:

$$v_{\text{jet}} \approx 200 \text{ km/s}$$

$$\tau_{\text{jet}} \approx 10^3 \text{ yr}$$

$$L_{\text{jet}} = v_{\text{jet}} \cdot \tau_{\text{jet}} \approx 6.3 \times 10^{15} \text{ m} \approx 0.2 \text{ pc}$$

3. Key Values

Quantity	Formula	Value
P_orb	Spectroscopic	44 yr
a_orb	Kepler 3rd law	~70 AU
F_U_Bi_i	UQFF	-2.09×10 N
M_total	Mira + WD	~2 M?
v_jet	HST proper motion	~200 km/s
t_jet	Jet age	~10 yr

4. Physical Significance

R Aquarii provides UQFF calibration at the symbiotic nova (compact binary) scale intermediate between individual stellar objects (PAPER_351 TDE) and galactic nuclei. The 44-year orbital period sets the longest UQFF binary activation period in the dataset (vs. PAPER_347 Cen A 12.5-yr AGN cycle). The Kozima LENR coupling at the mass transfer interface raises the testable prediction that R Aquarii's nova outburst energetics deviate from standard accretion models by an LENR fraction, detectable in nuclear gamma-ray emission (e.g., INTEGRAL 511 keV line observations).

5. Deduplication Note

- **vs. PAPER_322 (R Aquarii in SOURCE122):** Earlier treatment computed the 5-frequency resonances; this paper adds full F_U_Bi_i with a_orb from Kepler's third law and directly compares with HST 2025 jet data.
 - **vs. PAPER_351 (ASASSN-14li):** Different system class (symbiotic binary vs. TDE); both include F_Kozima.
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6. Classification

Physics Territory: FIRST UQFF F_U_Bi_i for a symbiotic nova binary with Kepler a_orb

Scale: Stellar (70 AU binary, ~200 pc distance)

CP Implementation: RAquariiSymbioticBinaryFUBiCalculator (CondensedPhysics3.py, Session 96)

UQFF computed: Canonical UQFF buoyancy parameter $U_{\text{bi}} = \frac{[SSq]GM}{r\kappa} = 5.0\text{e-}40.576.67\text{e-}11\text{M/r}$; for solar parameters: $U_{\text{bi,Sun}} = 5.7\text{e-}46.67\text{e-}111.99\text{e}30/(6.96\text{e}8) = 1.47\text{e+}2 \text{ m/s}$.